Sharzil Hasnine

+61 451 645 516 | sharzilhasnine975@gmail.com | linkedin.com/in/sharzilhasnine | github.com/Shas-9
Portfolio: shas-9.github.io

Education

Bachelor of Computer Science (Advanced) Feb 2023 – Nov 2027
Adelaide University

Technical Skills

Languages: SQL, C++, Java, Typescript, Javascript, Python, C, Bash
Databases: MySQL, MongoDB, RDBMS schema design, query optimisation
Development Practices: Unit testing, debugging, root-cause analysis, CI/CD, technical documentation, change management (Git, Jira)
Tools and Software: Git, Linux, Docker, Jira

Technical Experience

Industry Project - Block Model Compression Algorithm - Maptek Aug 2025 - Oct 2025
Tech: *C++, Bash, Jira*

- Analysed existing 3D data processing limitations and designed a stream-oriented compression system to address performance and storage deficiencies, achieving 98.8% reduction in data size and a 2.6x speed improvement.
- Tested, debugged, and validated program correctness via a purpose-built testing framework, enabling systematic fault detection independent of implementation strategy.
- Wrote and maintained technical documentation and build procedures as part of a cross-platform CI pipeline, improving team velocity by 1.5x and reducing onboarding friction.

Technical Volunteer - Autonomous navigation - Adelaide Rover Team Oct 2024 - Feb 2025
Tech: *Python, C++, Ros2, Nav2, Gazebo, Groot*

- Analysed system requirements and designed a modular behavior tree for the rover's autonomous navigation pipeline, enabling structured decision-making across path planning, recovery, and goal execution.
- Simulated and validated navigation behaviour prior to deployment, identifying and correcting logic faults ahead of the Australian Rover Challenge.

Technical Projects

Daily Iftar Registration System Feb 2026 - Mar 2026
Tech: *Typescript, Node.js, MongoDB, RESTAPI, Docker*
Link: *aumsa.com.au*

- Designed and maintained database schemas and API contracts in a live production system used by 70+ daily users, evaluating and evolving system design to support rapid feature expansion without service disruption.
- Diagnosed and corrected a class of date-handling faults by re-architecting DST-safe, timezone-aware logic (Australia/Adelaide), eliminating recurring off-by-one scheduling errors.

Distributed Weather Feed Aggregator with Lamport Clocks Aug 2025 - Sep 2025
Tech: *Java, JUnit, RESTAPI*
Link: *github.com/Shas-9/Weather-Aggregation-Server*

- Engineered a multi-threaded aggregation server with persistent storage, crash recovery, and automated stale-data expiry, correcting reliability faults under concurrent workloads.
- Applied unit testing to verify causal-consistency guarantees from a custom Lamport logical clock implementation.

Extracurricular Experience

Secretary Nov 2024 – Present
Islamic Student Society of The University of Adelaide (ISSUA), now AUMSA

- Guide and support a 10+ member committee, writing structured meeting minutes and action-item documentation to ensure alignment and accountability.
- Report on organisational status to committee stakeholders and maintain organised knowledge repositories via Google Drive.